

Analysis First-Year Exam, August 2017, Part 1

Answer FOUR questions in detail. State carefully any results used without proof.

1. A and B are nonempty sets of real numbers such that: (i) $A \cup B = \mathbb{R}$ and (ii) $a < b$ whenever $a \in A$ and $b \in B$. Prove that there exists a unique $s \in \mathbb{R}$ such that $(-\infty, s) \subseteq A$ and $(s, \infty) \subseteq B$.
2. Let X be a metric space, of which A and B are compact subsets; define $\delta := \inf\{d(a, b) : a \in A, b \in B\}$. Prove that there exist $a_0 \in A$ and $b_0 \in B$ such that $d(a_0, b_0) = \delta$. Must a_0 and b_0 be unique? Proof or counterexample, as appropriate.
3. Let f be a surjective map from the precompact metric space X onto the metric space Y . Prove that if f is uniformly continuous then Y is also precompact. Show by example that if uniformly is dropped then Y can fail to be precompact.
4. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous. Can it be the case that for each $t \in \mathbb{R}$, the value $f(t)$ is irrational iff t itself is rational? If yes, give an example; if no, give a proof.
5. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be differentiable and assume that $f(t) \rightarrow \ell \in \mathbb{R}$ as $t \rightarrow \infty$. Prove that if $f'(t) \rightarrow L \in \mathbb{R}$ as $t \rightarrow \infty$ then $L = 0$.